

ABSTRACT

This project proposes a **Python-based modular framework** to recommend suitable plants and support farming decisions using agricultural data. The system analyzes soil and environmental conditions and manages agricultural knowledge to provide accurate recommendations.

- Python-based intelligent framework
- Crop recommendation using agricultural data
- Modular architecture
- Easy to extend with Machine Learning
- Supports better farming decisions

INTRODUCTION

Choosing the right crop is essential for improving agricultural productivity. This project uses **Python** to process agricultural data and provide intelligent plant recommendations through a modular decision support system.

- Improves crop selection
- Uses soil and weather information
- Python for data processing
- Modular and scalable design
- Supports future AI integration

OBJECTIVES

The objective of this project is to develop a **Python-based modular framework** that analyzes agricultural data and provides intelligent plant recommendations for effective farming decision support.

Key Points:

- **Develop** a modular Python framework for intelligent plant recommendation.
- **Analyze** soil, weather, and crop data using Python to support decision-making.
- **Provide** accurate recommendations through efficient data processing and agricultural knowledge management.

LITERATURE SURVEY

S.No	Authors	Paper Title	Methods	Dataset	Accuracy	Key Findings
1	Sawant et al. (2026)	IoT-Driven Smart Crop Recommendation	GNB, IoT, ML	Soil & environmental data	99.55%	Real-time crop recommendation using IoT and ML.
2	Shastri et al. (2025)	ML & XAI for Crop Recommendation	Gradient Boosting, LIME	2,200 samples, 22 crops	99.27%	Improves accuracy and recommendation transparency.
3	Abu Bakr et al. (2025)	Learning-Based Crop Recommendation	RF, SVM, KNN, LSTM, GPT-2	2,200 samples, 22 crops	99.55%	Provides accurate and interactive farming recommendations.
4	Afzal et al. (2025)	Soil-Based ML Crop Recommendation	RF, XGBoost, Extra Trees	2,200 samples, 22 crops	98%	Soil and climate data improve crop prediction.
5	Senapaty et al. (2024)	ML-Based Crop Decision Support	13 ML classifiers, SMOTE	1,480 records, 37 crops	100%	Supports effective crop selection and farming decisions.

PROBLEM STATEMENT

Agriculture is highly dependent on factors such as **soil quality, weather conditions, and crop requirements**. Farmers often find it difficult to select the most suitable crop because this information is scattered and environmental conditions change frequently. Traditional decision-making methods may not always provide accurate or timely recommendations.

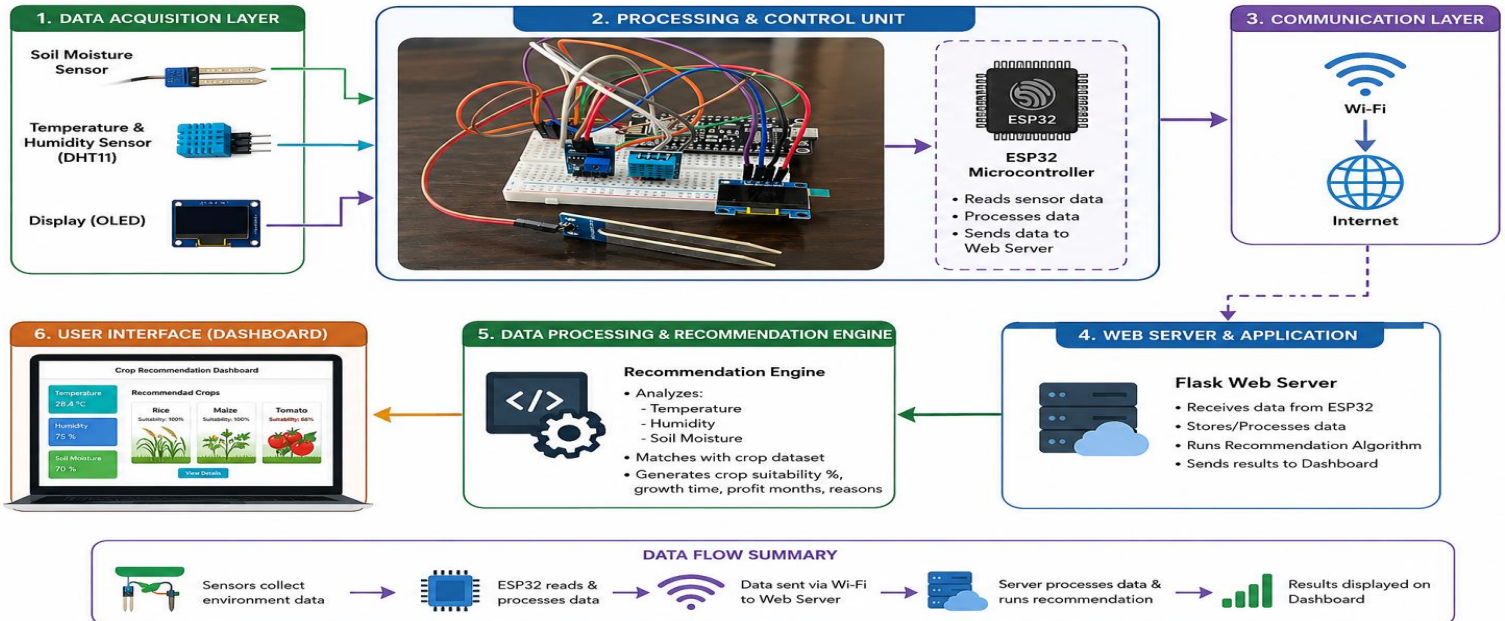
To address this challenge, this project proposes a **Python-based intelligent decision support framework** that processes agricultural data, manages crop knowledge, and generates reliable plant recommendations using a modular approach.

Key Points

- Difficulty in selecting suitable crops due to changing soil and weather conditions.
- Use Python for **data processing, analysis, and intelligent plant recommendation**.

SYSTEM ARCHITECTURE DIAGRAM

Crop Recommendation System



Module 1: Plant Suitability Analysis and Decision Support Engine

Definition

This module analyzes soil and environmental parameters to determine the most suitable plant for cultivation. Using **Python**, the system processes the input data, evaluates crop suitability, and generates intelligent recommendations to support farming decisions.

Key Functions

- Collect soil and environmental parameters (pH, temperature, rainfall, humidity, nutrients).
- Process and validate the input data using **Python**.
- Calculate plant suitability using decision-making algorithms.
- Rank and recommend the most suitable crops.
- Generate explainable recommendations for better decision support.

Module 1: Output



Smart Crop Recommendation System

Intelligent Farming Decision Support using Python & Flask

Temperature

28.0 °C

Humidity

75.0 %

Soil Moisture

70.0 %



Enter Farm Conditions

Temperature (°C)

Example: 28

Humidity (%)

Example: 75

Soil Moisture (%)

Example: 70

ANALYZE FARM CONDITIONS

Module 2: Agricultural Knowledge Repository and Data Processing Layer

Definition

This module serves as a centralized repository for agricultural information. Using **Python**, it stores, organizes, and processes crop-related data such as soil requirements, fertilizers, irrigation methods, and farming practices for efficient retrieval and analysis.

Key Functions

- Store and organize agricultural knowledge.
- Manage crop, soil, and fertilizer information.
- Process and retrieve agricultural data efficiently.
- Support Module 1 with structured and reliable data.
- Enable easy integration of new crops and farming information.

Module 2: Output

Maize

100.0% Suitable

✓ Temperature

Temperature is within the ideal range (18–30).

✓ Humidity

Humidity is within the ideal range (50–80).

✓ Soil Moisture

Soil moisture is within the ideal range (45–75).

 Growth Period
3–4 months

 Expected Yield
20–30 quintals/acre

 Cultivation Cost
₹28000/acre

 Suitable Season
Kharif / Rabi

★ **About this crop:** Maize grows well in warm conditions with moderate moisture.

★ **Highly Recommended**

Tomato

66.67% Suitable

✓ Temperature

Temperature is within the ideal range (18–30).

✗ Humidity

Humidity is too high. Ideal range is 50–70.

✓ Soil Moisture

Soil moisture is within the ideal range (40–70).

 Growth Period
2–3 months

 Expected Yield
80–120 quintals/acre

 Cultivation Cost
₹50000/acre

 Suitable Season
Year-round with suitable conditions

SOURCE CODE

Python

```
def calculate_suitability(temperature, humidity, soil_moisture, crop):  
  
    requirements = crop_data[crop]  
    score = 0  
  
    if is_in_range(temperature,  
                    requirements["temperature"][0],  
                    requirements["temperature"][1]):  
        score += 1  
  
    if is_in_range(humidity,  
                    requirements["humidity"][0],  
                    requirements["humidity"][1]):  
        score += 1  
  
    if is_in_range(soil_moisture,  
                    requirements["soil_moisture"][0],  
                    requirements["soil_moisture"][1]):  
        score += 1  
  
    return (score / 3) * 100
```

Python

```
def recommend_crop(temperature, humidity, soil_moisture):  
  
    results = []  
  
    for crop in crop_data:  
        suitability = calculate_suitability(  
            temperature,  
            humidity,  
            soil_moisture,  
            crop  
        )  
  
        results.append({  
            "crop": crop,  
            "suitability": round(suitability, 2)  
        })  
  
    results.sort(  
        key=lambda x: x["suitability"],  
        reverse=True  
    )  
  
    return results
```

ADVANTAGES

- Improves Crop Selection by recommending suitable plants based on agricultural conditions.
- Uses Soil and Weather Data such as pH, temperature, rainfall, humidity, and nutrients.
- Python-Based Processing enables efficient analysis and management of agricultural data.
- Modular Architecture makes the system easy to maintain, modify, and extend.
- Centralized Agricultural Knowledge provides organized crop, soil, fertilizer, and farming information.
- Supports Better Farming Decisions through ranked and explainable crop recommendations

FUTURE SCOPE

- Machine Learning Integration can improve crop prediction and recommendation accuracy.
- IoT Integration can collect real-time soil and environmental data from farms.
- Real-Time Weather Integration can provide recommendations based on changing weather conditions.
- Mobile/Web Application can make the recommendation system easily accessible to farmers.
- Support for More Crops can be added to expand the agricultural knowledge repository.
- Explainable AI can provide clearer reasons behind each crop recommendation

CONCLUSION

- The proposed system provides an intelligent approach to plant and crop recommendation.
- Python enables effective agricultural data processing and decision support.
- The system considers soil, weather, and crop-related information for recommendations.
- Its modular design allows easy expansion and integration of new features.
- The framework can help farmers make more informed and efficient crop-selection decisions.
- Future integration with Machine Learning, IoT, and AI can make the system more accurate and practical.

Reference

1. Sawant, Nitish Rupesh, et al. "An IoT-driven machine learning system for real-time smart crop recommendation and optimization in precision agriculture." *Discover Artificial Intelligence* 6.1 (2026): 194.
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3. Bakr, M. A., Khan, A. J., Khan, S. D., Zafar, M. H., Ullah, M., & Ullah, H. (2025). Evaluation of learning-based models for crop recommendation in smart agriculture. *Information*, 16(8), 632.
4. Afzal, Hadeeqa, et al. "Incorporating soil information with machine learning for crop recommendation to improve agricultural output." *Scientific reports* 15.1 (2025): 8560.
5. Senapaty, Murali Krishna, Abhishek Ray, and Neelamadhab Padhy. "A decision support system for crop recommendation using machine learning classification algorithms." *Agriculture* 14.8 (2024): 1256.